Amelie Norris

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Education

California State University, Monterey Bay — Monterey, CA
B.S. in Computer Science, 3.985 GPA — *Aug. 2024 – Dec. 2027*
- **Coursework:** Algorithms, Data Structures, Data Science, Database Systems, Discrete Math, Linear Algebra, Automata Theory; Break Through Tech / Cornell Tech ML Program (2025 – 2026)

Experience

Data Associate — Salinas, CA
Digital Nest — *Aug. 2026 – Present*
- Optimize data workflows in Airtable and Google Sheets, cleaning and processing records to maintain data integrity
- Reorganize the company's data structures with SQL and build interactive, user-friendly dashboards and visualizations for program reporting and annual grant reports

Machine Learning Engineer Intern — Remote
Anote, Inc. — *June 2026 – Aug. 2026*
- Extended the Synthetic Data Generator's Python pipelines to improve annotation accuracy and dataset quality; built automatable agent workflows for the AI-RFP product and resolved severe production bugs
- Built and evaluated LLM agent systems (function calling, structured outputs, system prompt design) with quantitative benchmarking of pipeline performance
- Served as technical point of contact on U.S. Department of Defense solicitations, authoring technical white papers and an SBIR Phase I technical volume for live SAM.gov RFPs

Backend Web Developer — Marina, CA
Claimzee — *Jan. 2026 – July 2026*
- Built API routes and server-side logic in **Next.js** and **TypeScript** for a production web application; helped secure the payment structure and fixed bugs found in QA before launch

Machine Learning Engineer, AI Studio Fellow — Remote
Anote, Inc. — *Aug. 2025 – Dec. 2025*
- Built an API-driven synthetic data system with a team of 5, generating labeled computer vision datasets (COCO, YOLO formats) with DALL·E 3, BlenderProc, and YOLO validation

Research & Publications

Research Fellow, Applied AI Research — Remote
Anote, Inc. — *2026*
- First author of an arXiv paper formulating posture and sustainment optimization as a finite-horizon **Markov Decision Process**; introduced the CEV optimizer and a RobustCEV extension that iterates against a Bayesian adversary
- Designed the experimental setup and evaluation, with baseline comparisons across three experiments in an Indo-Pacific basing environment, validated with paired t-tests and Bonferroni correction
- *Norris, A., Lee, A., Vidra, N., Setty, S. Posture and Sustainment Optimization Under Adversarial Uncertainty.* arXiv:2608.05256, 2026

Projects

Mona: Autonomous Options-Trading Engine | *Python, LLMs, Bayesian methods, Alpaca API* — 2026
- Built a system that converts unstructured analyst signals into structured views via an LLM layer; built a **Bayesian Black-Litterman** pricing layer; added rule-based risk management, paper-trading execution, and a React dashboard

Moosi | *Python, FastAPI, React, Tauri, libtorrent* — 2026
- Built a cross-platform desktop app (Tauri v2, React, Vite, FastAPI) with peer-to-peer file transfer via python-libtorrent, using MongoDB Atlas and Cloudflare R2 for metadata and object storage

Faultbench | *Python, LLM agents, evaluation* — 2026
- Built an adversarial evaluation-set generator for AI agents that produces stress-test cases surfacing failure modes standard benchmarks miss

Technical Skills

Languages: Python, Java, JavaScript/TypeScript, SQL, Rust, Lua, HTML/CSS
ML & Data: PyTorch, TensorFlow, Scikit-Learn, HuggingFace Transformers, OpenCV, pandas, NumPy, Matplotlib
Concepts: Multi-agent RL, LLM evaluation & benchmarking, RAG, synthetic data generation, prompt design, Bayesian inference
Frameworks & Tools: Flask, FastAPI, React, Next.js, Node.js, Git, AWS, GCP, MongoDB, PostgreSQL, Docker, Airtable, LaTeX

Leadership & Service

Resident Advisor, Student Housing and Residential Life, CSUMB — 2025 – Present
- Support a residential community with strict confidentiality and discretion; serve as RA on duty during on-call shifts, supporting students in emergencies

Science Student Success Peer Mentor, CSUMB — March 2025 – June 2026
- Mentor STEM students on coursework and career pathways